



# Commodore 128 — CPUs, Sound & I/

DIP pinouts from the top, notch up, pin 1 top-left. Power **red**, ground grey, clocks blue. Verify against the datasheet before powering up.

8502 · Z80 · 8580 SID · 6526 CIA · ALL DATASHEET-VERIFIED

## Legend — reading the pinouts notation

**/NAME Active LOW** — true / asserted when the pin is at logic **LOW** (~0 V). A leading / (datasheets often use an overline) marks these: e.g. /RES /IRQ /CS /WE.

**NAME Active HIGH / normal** — asserted when the pin is **HIGH** (~+5 V). No slash. R/W is a level: HIGH = read, LOW = write.

**Colour key:** **Vcc / Vdd = power** · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

## 8502 CPU (6510 family)

40-DIP

|            |          |    |      |
|------------|----------|----|------|
| φ0 in      | 1        | 40 | /RES |
| RDY        | 2        | 39 | R/W  |
| /IRQ       | 3        | 38 | D0   |
| /NMI       | 4        | 37 | D1   |
| AEC        | 5        | 36 | D2   |
| <b>+5V</b> | <b>6</b> | 35 | D3   |
| A0         | 7        | 34 | D4   |
| A1         | 8        | 33 | D5   |
| A2         | 9        | 32 | D6   |
| A3         | 10       | 31 | D7   |
| A4         | 11       | 30 | P0   |
| A5         | 12       | 29 | P1   |
| A6         | 13       | 28 | P2   |
| A7         | 14       | 27 | P3   |
| A8         | 15       | 26 | P4   |
| A9         | 16       | 25 | P5   |
| A10        | 17       | 24 | P6   |
| A11        | 18       | 23 | A15  |
| A12        | 19       | 22 | A14  |
| A13        | 20       | 21 | GND  |

C128 main CPU @ 1/2 MHz. 6510-family but with a 7-bit I/O port (P0-P6, pins 30-24) for tape/serial; note it has no φ2-out pin (R/W moves to 39, data bus to 38-31). Pinout verified from the MOS datasheet.

**Fails:** dead machine in 128/64 mode.

## Z80A CPU

40-DIP

|            |           |    |        |
|------------|-----------|----|--------|
| A11        | 1         | 40 | A10    |
| A12        | 2         | 39 | A9     |
| A13        | 3         | 38 | A8     |
| A14        | 4         | 37 | A7     |
| A15        | 5         | 36 | A6     |
| CLK        | 6         | 35 | A5     |
| D4         | 7         | 34 | A4     |
| D3         | 8         | 33 | A3     |
| D5         | 9         | 32 | A2     |
| D6         | 10        | 31 | A1     |
| <b>Vcc</b> | <b>11</b> | 30 | A0     |
| D2         | 12        | 29 | GND    |
| D7         | 13        | 28 | /RFSH  |
| D0         | 14        | 27 | /M1    |
| D1         | 15        | 26 | /RESET |
| /INT       | 16        | 25 | /BUSRQ |
| /NMI       | 17        | 24 | /WAIT  |
| /HALT      | 18        | 23 | /BUSAK |
| /MREQ      | 19        | 22 | /WR    |
| /IORQ      | 20        | 21 | /RD    |

Second CPU @ 4 MHz for CP/M mode (and the boot hand-off). The MMU selects which CPU drives the bus. **Fails:** no CP/M; can also stall boot.

## 8580 SID sound

28-DIP

|       |    |    |            |
|-------|----|----|------------|
| CAP1A | 1  | 28 | <b>Vdd</b> |
| CAP1B | 2  | 27 | AUDIO      |
| CAP2A | 3  | 26 | EXT IN     |
| CAP2B | 4  | 25 | <b>Vcc</b> |
| /RES  | 5  | 24 | POTX       |
| φ2    | 6  | 23 | POTY       |
| R/W   | 7  | 22 | D7         |
| /CS   | 8  | 21 | D6         |
| A0    | 9  | 20 | D5         |
| A1    | 10 | 19 | D4         |
| A2    | 11 | 18 | D3         |
| A3    | 12 | 17 | D2         |
| A4    | 13 | 16 | D1         |
| GND   | 14 | 15 | D0         |

8580 SID (or 6581 on early boards). Same pinout as the C64. **Fails:** no/weak sound, filter issues.

## 6526 CIA ×2

40-DIP

|            |           |    |       |
|------------|-----------|----|-------|
| Vss        | 1         | 40 | CNT   |
| PA0        | 2         | 39 | SP    |
| PA1        | 3         | 38 | RS0   |
| PA2        | 4         | 37 | RS1   |
| PA3        | 5         | 36 | RS2   |
| PA4        | 6         | 35 | RS3   |
| PA5        | 7         | 34 | /RES  |
| PA6        | 8         | 33 | D0    |
| PA7        | 9         | 32 | D1    |
| PB0        | 10        | 31 | D2    |
| PB1        | 11        | 30 | D3    |
| PB2        | 12        | 29 | D4    |
| PB3        | 13        | 28 | D5    |
| PB4        | 14        | 27 | D6    |
| PB5        | 15        | 26 | D7    |
| PB6        | 16        | 25 | φ2    |
| PB7        | 17        | 24 | /FLAG |
| /PC        | 18        | 23 | /CS   |
| TOD        | 19        | 22 | R/W   |
| <b>Vcc</b> | <b>20</b> | 21 | /IRQ  |

Two CIAs: CIA1 (\$DC00) keyboard/joystick, CIA2 (\$DD00) serial/user port. Same as C64. **Fails:** dead keyboard, no disk/IEC.

## Memory map

info

|                      |                                      |
|----------------------|--------------------------------------|
| <b>CPUs</b>          | 8502 (native) + Z80 (CP/M)           |
| <b>\$0000-\$FFFF</b> | banked RAM/ROM via 8722 MMU          |
| <b>Video</b>         | 40-col VIC-IIe + 80-col 8563 VDC     |
| <b>ROM</b>           | 32KB+ (BASIC 7.0, Kernal, C64, char) |

All four chips here are datasheet-verified. The video/custom ICs are on the companion sheet.

## Test & Repair

tips

|                          |                            |
|--------------------------|----------------------------|
| <b>Dead in all modes</b> | 8502 / 8722 MMU / 8721 PLA |
| <b>No sound</b>          | 8580 SID                   |
| <b>Dead keys / disk</b>  | 6526 CIA                   |
| <b>No CP/M</b>           | Z80 / MMU                  |

SID and CIA mirror the C64. A dead-in-every-mode machine points at the 8502, MMU, or PLA.